

1. Input

(1) 3DoFs: accelerometers; gyroscopes

(2) 追踪器性能指标

准确度 (Accuracy): The **similarity** between the measured value and the actual position; the further away from the origin of the reference coordinate system, the lower the accuracy.

抖动 (Jitter): The random noise in the tracker output, even when the tracked object remains stationary.

漂移 (Drift): The phenomenon where error accumulation and increase over time.

延迟 (Latency): The time delay between the moment of action and the system's output of the signal.

(3) Types of 3D Trackers

Mechanical Trackers: (adv) Accurate; small latency; no line-of-sight restriction (disadv) Interfere with the user's movement

Magnetic Trackers: transmitter generate magnetic field; the receiver senses the changes of magnetic field to calculate the position.
(AC: Eddy Currents issue) (DC: use Pulse emission mode)

Ultrasonic Trackers: Distance measurement using the flight time of sound waves, and use it to calculate the position (adv: price low; Not affected by metals) (disadv: line-of-sight restriction; affected by temperature, environmental noise; low update rate)

Optical Trackers: (Outside Looking In: The external camera captures the marks on the user's body) (Inside Looking Out: The sensor is on the user and marked in the environment) (adv: high accurate) (disadv: Occlusion marks on body parts cannot be traced.; ambiguity)

Inertial Trackers (gyroscope: detect angular velocity; accelerometer: detect linear acceleration; magnetometer: detect the intensity of Earth's magnetic field) (adv) no line-of-sight restriction; small/cheap/fast; low jitter (disadv) use accelerometer, has drift issue

(4) Gesture Devices

Pinch Gloves; 5DT Data Gloves; Didjigloves; CyberGloves

2. Output

(1) 视网膜生理结构

Rods: night vision

Cones: color vision, L/M/S cones 对 red/green/blue 敏感「绿色最敏感」。

(2) FOV

Each eye: 约 150° horizontal, 135° vertical。

Both eyes combined: 约 180° horizontal。

Binocular overlap: 约 120° horizontal。

The brain relies on the horizontal offset of images in overlapping areas to determine depth

(3) Depth Cues

Near object cues: convergence[eyes turn in] \ accommodation[lens change shapes] \ Disparity[two different views] \ Motion Parallax[fast vs. slow movement]

Far object cues: occlusion \ perspective \ haze \ **surface texture**

(4) Convergence-accommodation conflict

Eyes converge to virtual object depth but accommodate to the fixed display/virtual image plane. This mismatch causes discomfort.

(5) Sound Localization

方位角: ITD (Interaural Time Difference 双耳时间差) & IID (Interaural Intensity Difference, 双耳强度差); 低频 only by ITD

仰角: **Different frequencies are caused by the physical reflection of the pinna**

Distance Cues: Loudness/prior knowledge & motion parallax

(6) HRTF: Perform a transform on the relationship between the sound emitted by the sound source and the sound reaching the ear

(7) Haptic Feedback

Tactile feedback: vibration, texture, **pressure**, temperature

Force feedback: **weight**, resistance, collision

(8) CAVE 系统和多投影穹顶 (Domes) 为了产生无缝的大图像, 必须使用哪两种关键处理技术?

Blending (边缘融合) 处理重叠像素亮度 —— Pre-distort (预畸变) 矫正曲面投影变形

3. Architecture

3.1 流水线架构

(1) CPU & GPU

Cpu: Control-heavy tasks; space for Control & Cache

Gpu: Data-heavy applications \ Lots of parallelism; space for ALU

(2) Multi-pipeline Synchronization

Software Synchronization (软件同步)

Frame Buffer Synchronization (帧缓冲区同步)

Video Synchronization (视频同步)

3.2 分布式虚拟环境

(1) DVE (Distributed Virtual Environment)

(2) CVE (Collaborative Virtual Environment: A specific DVE that supports multiple users to complete tasks together in a virtual world)

(3) 为什么 DVE 难

User hope instant response

Single server has to handle all clients、updates objects、distribute messages

More users lead to server overloaded

(4) Multi-server architectures

User partitioning: users assigned to different servers

Region partitioning: VE divided into fixed regions

Dynamic partitioning: regions adjusted at runtime

(5) 同步解决方案

Causal ordering (因果排序): Events should be executed in the order they occur; only for discrete events

Roll-back (回滚机制):

Object locking (对象锁定)

Dead Reckoning: Both transmitter and receiver use motion predictor for per motion-object {predictor predict position->set a error threshold value->only error<et value send update information:save network bandwidth}

(6) DVE 可连接:virtual museum\online games\remote education

4.Application

(1)VR for medical(virtual anatomy; the surgical process; DRE trainer; MIS)

Train medical students and doctors

Reduce cost

Support repeated practice and assignment

Allow errors in virtual patients not real one

(2) Manufacturing Applications

VADE (Virtual Assembly Design Environment) : 验证 CAD 设计装配

Jack: 用于人机工程学分析的智能代理类人机器人 (humanoid agent)。

车身公差验证 (Car Body Tolerances)